

Maximal Margin Classifier

(do not include author names in document, as the lesson should be blinded for review)

Overview of Lesson

In this lecture students will learn Maximal Margin Classifier and shown the construction of MMC.

Type of Data

- dataset1 = [0, 0, 0, 1, 1, 1, 1, 1, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0, -1, -1, -1, -1, -1, 0]
- dataset2 = [0, 0, 0, 1, 1, 1, 1, 1, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 1, -1, -1, -1, -1, 0]

Learning Objectives

- Understand the concept of a hyperplane for data classification.
- Define and explain the Maximal Margin Classifier (MMC)
- Learn how to maximize the margin for optimal classification
- Apply the MMC to a dataset and find the optimal separating hyperplane.
- Discuss the impact of scaling on the margin and classification
- Recognize the limitations of MMC with non-linearly separable data.

Audience

- Undergraduate students who had taken calculus sequence

Time Required

90 mins suggestion with 1 break

Technology and Other Materials

- *Technology*: Students need their computers to complete the last part

Lesson Plan

This lesson was developed to give students an understanding of how the Maximal Margin Classifier works, particularly in the context of linear separation. The motivation comes from the widespread use of classifiers in machine learning, and understanding MMC provides a foundation for learning more advanced techniques such as Support Vector Machines (SVM). The approach is designed to allow students to develop both an intuitive and mathematical understanding of the concepts while also engaging them with practical examples.

First Stage of the Lesson- Setting the Stage (Instructor Led Introduction)

Start with motivation: Imagine you're trying to categorize emails as 'spam' or 'not spam.' Each email is represented by a set of features like word frequencies, sender, and time of day. The challenge is to find a decision rule that best separates the two categories.

- Motivation question 1: What does it mean to separate data points? How do we draw a line (or decision boundary) that divides different classes?
- Motivation question 2: If points are in 3D, what is the shape of 'separating line'?

Give definition of hyperplane:

In a p-dimensional space, a hyperplane is a flat affine subspace of hyperplane dimension $p - 1$.

$$X = \left\{ \left(x_1, \dots, x_p \right) \left| 0 + \beta_1 x_1 + \dots + \beta_p x_p = 0 \right. \right\}$$

We can separate the space into two pieces like $X > 0$ and $X < 0$. This will be a good method for classification of data set with 2 classes.

Second Stage of the Lesson – Participatory Learning (Small Group Work)

Question 1: What is the hyperplane for the dimension of dataset1?

Question 2: Find a hyperplane that separates the dataset1 (provided above)?

Question 3: How many hyperplanes you can use to separate dataset1? What can be the way to choose optimal one?

Introduce Maximal Margin Classifier:

Now suppose that we have an $n \times p$ data matrix X that consists of n training observations in p -dimensional space

$$D = \left\{ (x_i, y_i) \left| x_i \in R^p, y_i \in \{-1, 1\}, i \in \{1, \dots, n\} \right. \right\}$$

Our aim to find a hyperplane X so that it will be 'optimal'.

Our prediction function will be

$$p(x) = \beta^T x$$

Where $x = (1, x_1, \dots, x_p)$ and $\beta = (\beta_0, \dots, \beta_p)$

If $p(x)=y$ is greater than 0 we will assign 1, if not it will be -1.

Observe that $p(x)>0$ for $y=1$, and $p(x)<0$ for $y=-1$

So problem can be rewritten as

$$yp(x) > 0$$

Optimal: We will define the optimal one is the one that is farthest from training points.
Let M be the smallest distance from hyperplane to training set. Optimization problem can be rewritten as

$$\begin{aligned} & \text{maximize } M \\ & \sum_{i=1}^p \beta_i^2 = 1 \\ & y\beta^T x > 0 \end{aligned}$$

Discussion activity:

1. Working with a partner. Assume we are working in 2 dimension. Observe that $\beta = (\beta_0, \beta_1, \beta_2)$ and $x = (1, x, y)$ then it represents $\beta_0 + \beta_1 x + \beta_2 y = 0$. So $(\beta_1, \beta_2 \dots \beta_n)$ will determine the direction vector for a line. Discuss what are the meanings of β_0 , β_1 and β_2 ?
2. What is the difference scaling all of β 's and scaling only β_1 and β_2 ?
3. Can we use the β to optimize M?
 - a. Hint: The distance of x to hyperplane is

$$\frac{|\beta^T x|}{|\beta|}$$

What does the teacher do while students are working?

- Teacher is expected to walk around to see whether students are able to conduct a discussion. If not they can intervene and graph an example where beta is chosen to be (1,1,1) and (2,1,1). Comparing two graphs will show students a path to understand the first problem

Third Stage of the Lesson – Wrapping up the lesson (Whole Class Discussion)

Conclude the discussion and explain the reasoning behind the second condition

$$\frac{|\beta^T x|}{|\beta|} = \frac{|\beta^T x|}{\sum_{i=1}^p \beta_i^2}$$

So the margin will be computed as a distance from the point closest to

$$M = \frac{|\beta^T x|}{\sum_{i=1}^p \beta_i^2}$$

To make sure we are not affected by scaling of direction vector, we need to put the condition.

Hand-on activity: Please use your computer.

Find the optimal margin for the dataset1

Answer the following questions:

1. What is the downside of this method?
2. Will this method work well on dataset2?

References

James, G., Witten, D., Hastie, T., Tibshirani, R., & Taylor, J. (2023). *An introduction to statistical learning: With applications in Python*. Springer.